

Generative AI & AI Engineering Bootcamp (45 Hours)

Focus Areas: RAG Systems, AI Agents, LLM Applications, Prompt Engineering, APIs, and Production AI Engineering

Course Overview

This beginner-friendly course is designed specifically for aspiring AI Engineers. Unlike traditional ML-heavy courses, this program focuses heavily on modern Generative AI systems, Retrieval-Augmented Generation (RAG), AI Agents, LLM applications, APIs, and deployment workflows.

Module 1: AI Industry & Engineering Foundations (3 Hours)

- What is AI, ML, Deep Learning, and Generative AI
- Role of an AI Engineer
- How ChatGPT-like systems work
- Backend and frontend architecture overview
- Git, GitHub, Linux basics
- VS Code setup and virtual environments
- Open-source vs closed-source models

Module 2: Python for AI Engineering (6 Hours)

- Python syntax and data types
- Functions, loops, conditions
- Lists, dictionaries, tuples, sets
- File handling and JSON
- Exception handling
- OOP basics
- API handling using requests
- Working with environment variables

Module 3: Data Handling & NLP Basics (4 Hours)

- NumPy and Pandas basics
- CSV and JSON handling
- Text preprocessing
- Tokenization
- TF-IDF and embeddings overview
- EDA basics for text datasets

Module 4: Deep Learning & Transformer Foundations (5 Hours)

- ANN basics
- RNN, LSTM, BiLSTM
- Attention mechanism

- Transformer architecture
- Encoder vs Decoder
- Self-attention
- Why transformers dominate modern NLP

Module 5: LLMs & Prompt Engineering (5 Hours)

- What is an LLM
- How LLMs generate text
- Tokens and context windows
- Prompt Engineering
- Few-shot and Chain-of-thought prompting
- OpenAI, Gemini, Claude overview
- Open-source LLMs

Module 6: RAG Systems (8 Hours) — Core Focus

- What is RAG and why it matters
- Hallucination problem
- Chunking strategies
- Embeddings and semantic search
- Vector databases: ChromaDB, FAISS, Elasticsearch
- Document loaders and retrievers
- LangChain RAG pipeline
- Hybrid Search
- Parent Document Retriever
- Reranking and RRF overview
- Building PDF Chatbot
- Building company knowledge assistant

Module 7: AI Agents & Agentic Systems (9 Hours) — Core Focus

- What are AI Agents
- Tool Calling
- Memory systems
- Agent loops and reasoning
- Multi-agent systems
- Planning and execution
- LangGraph fundamentals
- CrewAI overview
- SQL Agents
- Research Agents
- Data Analysis Agents
- RAG + Agent integration
- Agent workflows

- Autonomous task execution
- Observability and monitoring

Module 8: APIs, FastAPI & Deployment (3 Hours)

- FastAPI basics
- Creating AI APIs
- Docker basics
- Environment variables
- Model deployment overview
- Production AI workflows

Module 9: Final Projects (2 Hours Guidance + Practice)

- RAG-based PDF Chatbot
- AI Research Assistant
- Multi-tool AI Agent
- Custom Knowledge Base Assistant
- AI Coding Assistant

Software Requirements

- Python 3.10+
- VS Code or Jupyter Notebook
- Git & GitHub
- Ollama for local LLMs
- CUDA-enabled GPU recommended but optional

Essential Libraries

pip install numpy pandas matplotlib scikit-learn openai transformers torch langchain chromadb faiss-cpu fastapi uvicorn sentence-transformers python-dotenv

Learning Outcomes

- Build AI chatbots using LLM APIs
- Create RAG systems with vector databases
- Develop AI Agents with tools and memory
- Understand transformer and LLM architecture
- Deploy AI applications using FastAPI
- Build production-ready AI engineering projects

Tutor Information

Tutor: Abinash Pant (AI/ML Engineer) at Leapfrog Technology